

ReGround

TEAM REPORT | HACKATHON CONCEPT AND NEW ZEALAND CONTEXT | 3 OCTOBER 2026

Planning-stage soil reuse & on-site conformance verification

Core proposition

Move soil reuse decisions upstream: use planning-stage earthworks and geotechnical information to identify likely surplus material, secure a suitable receiver before excavation, then verify the actual material on site against the receiver-defined acceptance envelope before direct haulage.

Executive summary

ReGround has evolved from a stand-alone soil separator into an integrated planning-and-verification pipeline. The core opportunity is not simply to find a use for excavated soil after it appears. It is to prepare the material pathway before excavation, so that suitable surplus material can move directly from source to a pre-arranged local receiver when on-site checks confirm conformance.

The approach combines three layers: (1) planning-stage material forecasting and receiver coordination, (2) existing professional and regulatory evidence, and (3) a mobile physical unit that segregates and continuously checks the excavated stream for agreed properties such as particle-size fraction, moisture, mass and visual consistency. ReGround does not replace contamination testing, geotechnical design or receiver approval.

The problem ReGround targets

Current friction	Why it matters for reuse
Surplus is forecast early, but final destinations can remain unresolved until detailed design / contractor planning.	Late routing increases the risk of stockpiling, rehandling, long-distance disposal or losing a reuse opportunity.
Receiver requirements vary by intended use, consent and operating criteria.	A “match” is not an approval; evidence must be assembled for the specific receiver.
Soil is heterogeneous and actual excavation conditions can differ from forecast.	A receiver needs confidence that the delivered stream still matches the material it reviewed.
Transport and disposal are highly location- and programme-dependent.	A nearby, timed match can create both environmental and commercial value.

Recommended hackathon framing

One region + one material stream + one receiver market. Demonstrate the full chain from planned surplus → receiver requirements → conditional approval → on-site verification → direct local reuse. Keep contamination, legal approval and engineering suitability with the relevant professionals and receiving party.

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1. How surplus soil moves through a New Zealand earthworks project

1. Current earthworks workflow

The workflow below is a practical synthesis of NZTA earthworks practice, WasteMINZ surplus-soil guidance and receiving-site requirements. Exact responsibilities vary by project and contract, but the sequence is useful for locating the intervention point.

Stage	What is happening
1. Site investigation	Geotechnical / environmental information, site history, likely soil layers.
2. Earthworks design	Proposed levels, cut and fill quantities, earthworks methodology.
3. Cut / fill balance	Estimate how much excavated material can be reused on site and what may remain surplus.
4. Material suitability	Identify material suitable for structural or other on-site uses; separate unsuitable / excess material.
5. Surplus forecast	Expected material type, volume, timing and production rate become visible.
6. Off-site pathway	Internal reuse, another project, reuse facility, managed fill / cleanfill, or disposal.
7. Receiver review	Receiver checks its own WAC / consent / operating and commercial requirements.
8. Approval / agreement	Quote, acceptance conditions, delivery window and logistics are agreed.
9. Excavation	Actual material is exposed; variability and field condition become known.
10. Verification + haulage	Material is checked / segregated and sent to the approved route.

Evidence that surplus is visible before excavation

NZTA's Warkworth-to-Wellsford AEE quantified 12.35 million m³ of total cut, 9.613 million m³ of fill, 7.182 million m³ of cut considered suitable for structural fill, and about 3.379 million m³ of excess soil for disposal. The report also states that final disposal locations and volumes would be determined during detailed design once a contractor and construction methodology were confirmed. This illustrates the planning gap ReGround targets: surplus can be forecast well before the final route is locked in. [5]

Key interpretation

ReGround does not need to invent a new point in the project lifecycle. It moves receiver discovery, evidence gathering and conditional acceptance as far upstream as practical, while leaving a fast field verification step on the construction critical path.

2. Regulatory and acceptance pathway - what ReGround can and cannot do

2. Regulation, guidance and receiver acceptance are different layers

Layer	Examples	Implication for ReGround
Statutory / national rules	NESCS for certain soil disturbance on HAIL land; Waste Minimisation Act levy / reporting framework.	The platform cannot grant consent or declare contaminated soil safe.
Council / plan rules	Auckland Unitary Plan land-disturbance rules; E13 cleanfill / managed-fill / landfill provisions and site-specific resource consents.	Requirements depend on source and receiving location.
Industry guidance	WasteMINZ Technical Guidelines: Characterising Surplus Soil for Disposal (Version 2.0, Sep 2024).	Provides good-practice sampling / information workflow, but is not statutory approval.
Engineering specification	NZTA F/1 and project-specific earthworks / geotechnical specifications.	Suitability for engineered use remains a professional / project decision.
Receiver acceptance criteria	Facility-specific WAC, testing, moisture / compaction requirements, operating limits.	Final acceptance belongs to the receiving operator.

WasteMINZ responsibilities most relevant to the concept

- Producers are responsible for collecting information on surplus soil; the guidance says this should ideally occur in advance of engaging contractors so costs and alternatives can be considered. [1]
- Receivers are responsible for requesting enough characterisation information to address WAC and resource-consent compliance. [1]
- Before transport, the producer provides sufficient information for a quote and approval; the receiving operator ultimately accepts or rejects the material. [1]
- Direct engagement with the proposed receiver is required because receiving sites differ in consent obligations, risk profile and operating requirements. [1]
- WasteMINZ sets a baseline of 1 sample per 25 m³ with a minimum of 3 per consignment for the guidance’s typical smaller-volume context; larger projects may use other SQEP-agreed methods. [1]

What is known before excavation vs what remains uncertain

Usually available before excavation	Usually confirmed / variable during excavation
Source address, site history / HAIL context	Actual moisture at time of excavation
Cut/fill geometry and expected volume	Actual fraction yield after screening
Borehole / test-pit soil layers and geotechnical evidence	Short-range heterogeneity and unexpected material
Existing laboratory results / contamination evidence	Actual daily production and weather effects
Construction programme and expected delivery window	Whether each batch continues to conform to the pre-approved profile
Receiver-specific requirements once a receiver is engaged	Hold / retest / reroute decisions for non-conforming batches

Design principle

Pre-characterise before excavation. Pre-agree the acceptance envelope. Verify conformance on site. Haul directly when the batch passes.

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3. ReGround operating model - from planning record to direct haulage

3. Proposed ReGround workflow

Phase	ReGround role	Output
A. Planning	Ingest cut/fill forecast, geotech / environmental evidence, expected material and programme.	Planned material record.
B. Market / receiver coordination	Search nearby demand or receiver; compare material, volume, timing, distance and required evidence.	Potential local route.
C. Pre-acceptance	Expose receiver-specific requirements; complete missing tests / professional review outside the excavation critical path.	Conditional acceptance / offtake conditions.
D. Excavation	Mobile unit screens / segregates the stream and records mass, moisture, particle fraction and visual consistency.	Actual batch record.
E. Conformance gate	Compare batch against receiver-defined acceptance envelope.	PASS / HOLD / FAIL.
F. Logistics	PASS batches are loaded and hauled directly; other batches are isolated for retest or alternative routing.	Reduced stockpile / rehandling and avoidable disposal.

Conventional pathway vs ReGround

Conventional late-path risk	ReGround approach
Surplus identified; destination may still be unresolved.	Potential receiver is identified from the planning-stage forecast.
Evidence is assembled after routing decisions become urgent.	Receiver requirements are known early; missing evidence is completed before earthworks.
Excavated material may wait in stockpile while acceptance / logistics are arranged.	Only field conformance verification remains after excavation begins.
Sampling / inspection observes only parts of the stream.	The mobile unit can continuously observe selected physical attributes across the processed stream.
Material may default to a disposal route if time / storage is constrained.	Passing batches can move directly to a pre-arranged local reuse route.

What the physical unit is - and is not

On-site conformance verification

The unit is a configurable QA gate, not a universal soil-certification machine. It checks whether the excavated stream remains consistent with the pre-characterised material and the receiver's agreed physical envelope. Contamination certification, geotechnical design and legal approval remain external.

Example configurable fields: source zone / batch ID; mass; moisture range; oversize limit; coarse / intermediate / fines fraction; visual material class or anomaly flag; required evidence reference. The exact profile should be set by the chosen receiver / use case.

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4. Alternatives and comparable services - where they intervene

4. Existing ecosystem and alternatives

Alternative	Structure	Where it intervenes	What it solves	Gap relative to ReGround
Springline (NZ)	Environmental / soil-management consultancy	Planning → construction	Characterisation, reuse / disposal pathways, regulatory navigation, stakeholder engagement.	Expert service rather than an integrated two-sided market + field verification system.
CivilShare (NZ)	External civil-contractor marketplace	After material / resource is available or listable	Buy, sell, share, swap and hire materials / resources; "dirt to diggers".	Broad marketplace; does not itself provide receiver-specific soil approval or conformance verification.
Mutu (NZ/AU)	Internal resource-sharing SaaS	Across projects within one business	Lists, searches, reserves and transfers surplus materials and equipment; tracks savings and waste diversion.	Strong internal visibility, but not an external soil acceptance / offtake workflow.
BRANZ Resource Recovery Map (NZ)	Public directory / map	Destination discovery	Finds local reuse, recycling and disposal facilities by material type / location.	Directory only; no trade, technical approval or batch verification.
Cleanfill / managed-fill operators	Receiving facility	Receiver approval + disposal / fill	Apply site-specific acceptance criteria, pre-approval, operational limits and gate fees.	Endpoint rather than market coordination layer.
Earth Exchange (UK)	Map-based soil / aggregate exchange + MMP service	Planning → exchange → project completion	Matches surplus and demand; offers fully managed Materials Management Plan support.	Closest analogue; UK regulatory context and no ReGround-style physical field-verification layer.
Bulk Exchange (US)	Heavy-civil marketplace / procurement platform	Procurement and disposal planning	Supplier / disposal discovery, pricing and bid connection for aggregates, sand, soil and disposal sites.	Broader procurement platform; not specific to NZ soil compliance / on-site conformance.

Positioning hypothesis

ReGround is not "another marketplace".

The differentiated position is an end-to-end soil-reuse readiness pipeline: future surplus forecast → receiver-specific evidence / conditional offtake → excavation-stage segregation and conformance verification → direct local delivery.

The closest precedents validate individual pieces of the model. Springline validates the need for early soil-pathway expertise; CivilShare / Earth Exchange validate material exchange; Mutu validates the value of cross-project visibility; receiving sites validate pre-approval and batch acceptance. ReGround's hypothesis is that integrating those steps earlier reduces critical-path friction.

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5. Environmental and economic value - how the benefit is created

5. Environmental mechanism

Mechanism	Environmental effect
Earlier local match	A receiver is secured before soil enters the disposal chain.
Less disposal	More suitable soil is diverted from cleanfill / landfill / managed-fill routes.
Shorter haulage	Source-to-receiver distance can be shorter than source-to-disposal distance.
Potential virgin-material displacement	If the receiver would otherwise import fill / soil / aggregate, reuse can also avoid an inbound material movement and some extraction / processing.
Less rehandling	Direct routing can reduce stockpile loading / unloading and associated plant activity.

Core measurement equations

Transport

Avoided tonne-km = material mass × (baseline outbound disposal distance + avoided inbound virgin-material distance – direct reuse distance).

Disposal

Potential disposal diversion = tonnes of the matched stream that are actually accepted and reused ÷ tonnes of that stream generated.

Carbon

Estimated transport CO₂e can be calculated later by multiplying avoided tonne-km by an appropriate freight emissions factor. Keep the factor and system boundary explicit.

Economic mechanism

Potential saving	How it arises
Disposal / gate fees	Material is reused rather than sent to a charging disposal facility.
Haulage	Shorter route and potentially fewer loaded-material movements.
Rehandling / stockpile	Less double handling while waiting for a destination or approval.
Programme risk	Technical review and receiver coordination are shifted before excavation.
Virgin material procurement	Receiver may avoid buying / importing equivalent suitable material.
Less value: ReGround costs	Mobilisation, screening / power, professional testing, additional handling, platform / service and transport to receiver must be deducted.

Concrete NZ cost signal

MfE levy rates from 1 July 2026 are NZ\$70/t for Class 1 municipal disposal facilities, NZ\$40/t for Class 2 C&D facilities and NZ\$15/t for Class 3/4 managed / controlled fills; Class 5 cleanfills are not levied. These are levy rates, not total gate fees. [6] Goodman Contractors' Silverstream Cleanfill currently lists a NZ\$27/t cleanfill rate, subject to strict pre-approval and acceptance criteria. [13]

Illustrative only: at a facility charging \$27/t, avoiding disposal of 1,000 t could avoid up to \$27,000 of that gate charge before subtracting ReGround, testing, receiver and transport costs. The business case must use the actual source, receiver and quotes chosen for the hackathon case.

6. Time: the main gain is moving work off the excavation critical path

No authoritative NZ-wide median was found for seller-to-receiver surplus-soil approval time. The practical duration depends on whether a receiver is already known, whether the evidence package is complete, and whether new sampling, SQEP review or consent work is required. For the hackathon, any timing numbers should be labelled as scenario assumptions rather than industry averages.

Scenario	Illustrative planning assumption*	Critical-path implication
Known receiver + complete evidence	~1–3 working days	Mostly booking / review / logistics.
New receiver + existing evidence	~3–7 working days	Receiver review, commercial terms and scheduling.
Additional sampling / laboratory evidence	~1–2 weeks	Can delay routing if started only after excavation.
Complex DSI / consent / specialist case	~2–6+ weeks	Must remain outside ReGround's "fast verification" claim.

**Illustrative case-study assumptions only; not published NZ median transaction times.*

What ReGround should claim

Do claim	Do not claim
Moves receiver coordination and missing-evidence work earlier.	Automatically certifies soil as safe or legally reusable.
Can compare actual batch properties with a pre-agreed acceptance profile.	Replaces SQEP / laboratory contamination testing.
Can reduce post-excavation routing delay when a pre-arranged receiver exists.	Guarantees a fixed number of days saved on every project.
Can provide continuous / batch-level physical data across the processed stream.	Proves superior accuracy to professional sampling without validation data.

Hackathon proof structure

- One source project with a plausible cut/fill forecast and expected surplus stream.
- One selected receiver / buyer use and a transparent acceptance envelope.
- One pre-excavation evidence package: planned material + volume + timing + required tests / approvals.
- One Blender / digital physical demonstration of separation and field conformance verification.
- One PASS / HOLD / FAIL example tied to a receiver-defined profile.
- One baseline-vs-ReGround comparison using disposal tonnes, tonne-km, gate / transport cost assumptions and planning lead time.

Open decisions for the team

Next decision

Choose the first receiver market and material stream. The acceptance profile should then drive sensor selection, the routing demo, the physical verification logic and the impact model.

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Sources and scope notes

7. Sources used for this update

1. WasteMINZ - Technical Guidelines: Characterising Surplus Soil for Disposal, Version 2.0 (Sep 2024). [Link](#)
2. Ministry for the Environment - National Environmental Standard for Assessing and Managing Contaminants in Soil to Protect Human Health (NESCS). [Link](#)
3. Auckland Unitary Plan - E13 Cleanfills, managed fills and landfills. [Link](#)
4. NZ Transport Agency - Earthworks construction specification TNZ F/1. [Link](#)
5. NZ Transport Agency - Warkworth to Wellsford Assessment of Effects on the Environment, Part Two (2020). [Link](#)
6. Ministry for the Environment - Waste disposal levy rates and reporting requirements. [Link](#)
7. Springline - Soil Management. [Link](#)
8. CivilShare - civil contractor marketplace. [Link](#)
9. Mutu - construction resource management / surplus materials. [Link](#)
10. BRANZ - Resource Recovery Map. [Link](#)
11. Earth Exchange UK - soil and aggregates exchange / MMP service. [Link](#)
12. Bulk Exchange - heavy construction marketplace. [Link](#)
13. Goodman Contractors - Silverstream Cleanfill acceptance requirements and current rate. [Link](#)

Scope and caution

This report is a hackathon framing document, not legal, environmental or geotechnical advice. Regulations, consent conditions and receiver acceptance criteria are site- and use-specific. ReGround should expose evidence status and required professional review rather than imply automatic compliance or approval.

The report deliberately avoids relying on broken or unverified links from earlier drafts. The sources above were re-checked as accessible during preparation on 3 October 2026.

Working pitch

ReGround

We move soil reuse preparation before excavation: forecast the surplus, secure a receiver and its acceptance conditions, then verify the actual stream on site so passing material can move directly to local reuse instead of defaulting to stockpile and disposal.